

# MOMENTUM HELICITY SEPARATES CASCADE OPERATORS FROM EULER–ARNOLD OPERATORS

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ABSTRACT. Paper III leaves open whether a local cascade operator in Tao’s sense can be an Euler–Arnold operator, and hands the question to real algebraic geometry. The obstruction there is structural: every argument of Paper I–Paper III is built on the energy identity, which is precisely the invariant Tao’s cancellation condition (4.3) is designed to preserve, so no linear functional can separate the two classes (Paper II, Corollary 5.4). Three-dimensional ideal flow carries a second structural invariant that nothing in (4.3) imposes: the helicity of the *momentum*  $m = Au$ , a Casimir of the Lie–Poisson structure and therefore conserved by every Euler–Arnold flow, for every inertia operator. We show that the resulting condition collapses to linear algebra. On  $k^\perp$  the curl acts as  $i|k|J$  with  $J$  a quarter turn,  $A^{1/2}$  is a symmetric  $2 \times 2$  matrix, and  $SJS = \det(S)J$  for symmetric  $S$ ; consequently the helicity defect is *linear in the numbers  $\sqrt{\det A_k}$  and sees the inertia operator through nothing else*. An operator is Euler–Arnold only if an explicit matrix  $M(C)$  has an entrywise positive null vector. Euler–Arnold vertices give corank exactly one, with that null vector; cascade operators give full rank, for two, three and four species, on five mode sets, and for Tao’s own coefficients the null vector exists but fails positivity. The species count never enters the criterion, which is why it reaches the case Paper III could not.

## 1. THE QUESTION, AND WHY THE TRILOGY COULD NOT ANSWER IT

Paper III ends at a dimension count. On a web of triads it measures two parametrised images, that of  $\{A_k\} \mapsto (T_A(\tau))_\tau$  and that of  $(\alpha, \{G_k\}) \mapsto ((G \otimes G \otimes G)\alpha)_\tau$ , finds a codimension growing quadratically in the number of legs, and stops, because two subvarieties whose dimensions sum to less than the ambient may still meet in special position.

The obstruction to going further is not the dimension count. It is that Paper I–Paper III are built on one invariant, and it is the wrong one. The Euler–Arnold class conserves energy; Tao’s condition (4.3) is imposed *precisely so that his averaged operator does too* [7, §4]; and Paper II, Theorem 5.1 shows the energy relations exhaust the linear information available on a triad. Corollary 5.4 then records the consequence: no linear functional vanishing on the Euler–Arnold class can exclude any cascade operator. Three papers were spent searching for a separator among functionals that every cascade satisfies by construction.

Three-dimensional ideal flow has a second structural invariant. For a divergence-free momentum one-form  $m$  on  $\mathbb{T}^3$  the helicity

$$H(m) = \int_{\mathbb{T}^3} m \wedge dm = \int_{\mathbb{T}^3} m \cdot \operatorname{curl} m$$

is a Casimir of the Lie–Poisson structure on  $\mathfrak{sdiff}(\mathbb{T}^3)^*$ ; see [4, §I.9], though Lemma 3.1 below proves what we need directly and in three lines. Casimirs do not depend on the Hamiltonian, so  $H(m)$  with  $m = Au$  is conserved by *every* Euler–Arnold flow, whatever the inertia operator; and nothing in (4.3) imposes it on a cascade. That asymmetry is the content of this paper.

The condition it yields is not a linear functional on the vertex — it involves  $A$  — so Corollary 5.4 does not reach it. It does not use invertibility of the frames, and the species count does not appear in it, so the  $m \geq 3$  wall of Paper III does not reach it either.

*Remark 1.1* (relation to the literature). Tao states the deficiency himself, and locates it in the vorticity formulation: the true nonlinearity admits one involving only differential operators, whereas a general averaged operator does not, so that any proof of global regularity must use structure of the nonlinearity that the averaged version does not share [7]. What is left implicit is *which* structure. This paper names one, and the naming is what makes it testable — the vorticity formulation is what carries the Casimirs of the Lie–Poisson structure, and the momentum helicity is the scalar that survives of them.

The complementary programme is well developed, and it is worth saying where this sits relative to it. The  $\alpha$ -models are designed to preserve exactly what Tao’s averaging discards: the Navier–Stokes- $\alpha$  model is obtained by filtering the velocity of the fluid loop in Kelvin’s circulation theorem [5], and the Lagrangian-averaged Euler equations are Euler–Poincaré equations on the same Lie algebra with a modified inertia operator [6] — that is, they are members of the class  $\{B_A\}$  studied here, with  $A = I - \alpha^2 \Delta$  in place of  $A = I$ . The two constructions bracket the question. Averaging that respects the circulation theorem stays inside the Euler–Arnold class by design; averaging of Tao’s local cascade type, as shown below, cannot enter it for any inertia operator whatsoever.

## 2. PRELIMINARIES

We keep the notation of Paper I. Let  $\mathfrak{g}$  be a Lie algebra with an inner product  $\langle \cdot, \cdot \rangle$  and  $A: \mathfrak{g} \rightarrow \mathfrak{g}$  symmetric and positive definite, with  $\text{ad}^*$  defined by  $\langle \text{ad}_x^* y, z \rangle = \langle y, [x, z] \rangle$ . In the energy-normalised variable  $X = A^{-1/2} m$ , the Euler–Arnold equation reads  $\dot{X} = B_A(X, X)$  with

$$B_A(X, Y) = \frac{1}{2} \left( A^{-1/2} \text{ad}_{A^{-1/2} X}^* (A^{1/2} Y) + A^{-1/2} \text{ad}_{A^{-1/2} Y}^* (A^{1/2} X) \right), \quad (1)$$

and  $S(x, y, z) := \langle y, [x, z] \rangle$ . Taking  $Y = X$  in (1) and writing

$$u := A^{-1/2} X, \quad m := A^{1/2} X, \quad \text{so that } m = Au, \quad (2)$$

gives  $\langle B_A(X, X), Z \rangle = S(u, m, A^{-1/2} Z)$ , that is

$$A^{1/2} B_A(X, X) = \text{ad}_u^* m = \dot{m}. \quad (3)$$

Throughout,  $\mathfrak{g}$  is the algebra of divergence-free, mean-zero vector fields on  $\mathbb{T}^3$ , and we work on a finite set  $\mathcal{S}$  of wavevectors closed under  $k \mapsto -k$ . Since the fields are real-valued, their Fourier coefficients satisfy  $\hat{v}(-k) = \overline{\hat{v}(k)}$ , and for  $A$  to preserve that we need  $A_{-k} = A_k$  as operators on  $k^\perp = (-k)^\perp$ , so  $A$  contributes one symmetric  $2 \times 2$  matrix per pair  $\{k, -k\}$ ; we write  $\mathcal{P}$  for the set of pairs.

**Definition 2.1.** A local cascade operator with  $m$  species is determined by constants  $\alpha \in \mathbb{R}^{m^3}$  and divergence-free profiles  $\psi_i(k) \in k^\perp$  through

$$\langle C(X, X), Z \rangle = \sum_{k+l+p=0} \sum_{i,j,q} \alpha_{ijq} \langle \psi_i(k), \hat{X}(k) \rangle \langle \psi_j(l), \hat{X}(l) \rangle \langle \psi_q(p), \hat{Z}(p) \rangle,$$

the sum over ordered triples from  $\mathcal{S}$ .

**Lemma 2.2.** Under Tao’s symmetry (4.2), his cancellation condition (4.3) is exactly the vanishing of the cyclic symmetrisation of  $\alpha$ ,

$$\alpha_{ijq} + \alpha_{jq i} + \alpha_{qi j} = 0 \quad \text{for all } i, j, q,$$

and this holds if and only if  $\langle C(X, X), X \rangle \equiv 0$ . Together with (4.2) itself the admissible  $\alpha$  therefore form a linear space of dimension  $\frac{1}{3}(m^3 - m)$ , namely 2, 8, 20 for  $m = 2, 3, 4$ . Dropping (4.2) and imposing only the cyclic condition doubles this to  $\frac{2}{3}(m^3 - m)$ ; the extra directions are those antisymmetric in the first two indices, and they contribute nothing, since the triad sum runs over ordered pairs and is therefore symmetric in those indices. They give the zero operator.

*Proof.* By (4.2),  $\alpha_{i_1 i_2 i_3} = \alpha_{i_2 i_1 i_3}$ , so the six permutations occurring in (4.3) coincide in pairs and their sum is twice the cyclic sum. Setting  $Z = X$  in Definition 2.1 and grouping the ordered triples  $(k, l, p)$  into cyclic classes, the coefficient of  $\langle \psi_i(k), \hat{X}(k) \rangle \langle \psi_j(l), \hat{X}(l) \rangle \langle \psi_q(p), \hat{X}(p) \rangle$  is  $\alpha_{ijq} + \alpha_{jq i} + \alpha_{qi j}$ ; for this to vanish for all profiles and all  $X$  is the stated condition. The dimension count is  $m^3$  less the number of cyclic orbits in  $\{1, \dots, m\}^3$ , which is  $m + (m^3 - m)/3$ .  $\square$

*Remark 2.3.* Lemma 2.2 is Paper II, Theorem 5.1 read on the coefficients rather than on the blocks, and it is the constraint that was omitted from the measurement in Paper III, Proposition 8.1; the corrected table there now uses it.

### 3. THE CASIMIR

**Lemma 3.1.** *For every  $A$  and every solution of the Euler–Arnold equation,  $\frac{d}{dt}H(m) = 0$ , where  $m = A^{1/2}X$  as in (2).*

*Proof.* It suffices that  $\langle m, [\text{curl } m, w] \rangle = 0$  for every  $w \in \mathfrak{g}$ , that is  $\text{ad}_{\text{curl } m}^* m = 0$ ; the conservation law then follows from (3) and  $\langle a, \text{curl } b \rangle = \langle \text{curl } a, b \rangle$ :

$$\frac{1}{2} \frac{d}{dt} H(m) = \langle \dot{m}, \text{curl } m \rangle = \langle \text{ad}_u^* m, \text{curl } m \rangle = \langle m, [u, \text{curl } m] \rangle = -\langle m, [\text{curl } m, u] \rangle = 0.$$

For the first claim write  $\omega = \text{curl } m$  and  $I = \int m_i (\omega_k \partial_k w_i - w_k \partial_k \omega_i)$ . Integrating the first term by parts and using  $\text{div } \omega = 0$  gives  $\int m_i \omega_k \partial_k w_i = -\int (\omega_k \partial_k m_i) w_i$ . For the second,  $\int m_i w_k \partial_k \omega_i = \int w_k \partial_k (m_i \omega_i) - \int w_k \omega_i \partial_k m_i$ , and the first piece is  $-\int (\text{div } w)(m \cdot \omega) = 0$ . Hence

$$I = \int w_k \omega_i (\partial_k m_i - \partial_i m_k) = \int w_k \omega_i \varepsilon_{kil} \omega_l = 0,$$

since  $\partial_k m_i - \partial_i m_k = \varepsilon_{kil} \omega_l$  and  $\varepsilon_{kil}$  is antisymmetric in  $(i, l)$ .  $\square$

The proof uses nothing about  $A$ ; it uses only that  $\dot{m} = \text{ad}_v^* m$  for *some*  $v$ . That is exactly why the condition has content for a cascade, whose  $\dot{m}$  need not be of that form.

**Definition 3.2.** For a bilinear operator  $C$  on  $\mathfrak{g}$  and a positive definite  $A$ , the *helicity defect* is

$$D(X; A) := \langle A^{1/2} C(X, X), \text{curl } (A^{1/2} X) \rangle = \frac{1}{2} \frac{d}{dt} H(m) \Big|_{\dot{X}=C(X, X)}.$$

By Lemma 3.1,  $D(X; A) \equiv 0$  whenever  $C = B_A$ .

### 4. THE REDUCTION TO DETERMINANTS

**Lemma 4.1.** *For every symmetric  $2 \times 2$  matrix  $S$  and  $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  one has  $SJS = \det(S) J$ .*

*Proof.* With  $S = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ ,  $SJ = \begin{pmatrix} b & -a \\ c & -b \end{pmatrix}$  and  $SJS = \begin{pmatrix} 0 & b^2 - ac \\ ac - b^2 & 0 \end{pmatrix} = (ac - b^2) J$ .  $\square$

The identity is special to two dimensions; in higher dimensions  $SJS$  is not a multiple of  $J$ . That  $k^\perp$  is two-dimensional is what makes the next proposition possible.

**Proposition 4.2.** *Fix an orthonormal frame of  $k^\perp$  for each pair, positively oriented with respect to  $k$ , and write  $\hat{C}(k), \hat{X}(k) \in \mathbb{R}^2 \otimes \mathbb{C}$  in that frame. Then*

$$D(X; A) = \sum_{k \in \mathcal{P}} \sqrt{\det A_k} c_k(X), \quad c_k(X) = -2|k| \operatorname{Im} \langle \hat{C}(k), J\hat{X}(k) \rangle. \quad (4)$$

*The inertia operator enters only through the numbers  $\sqrt{\det A_k}$ , one per leg pair, and enters linearly.*

*Proof.* On  $k^\perp$  the curl acts by  $a \mapsto ik \times a$ , and  $k \times$  is  $|k|$  times the quarter turn  $J$  in the chosen frame; equivalently  $|k|Ja = k \times a$ , so that  $c_k(X) = -2 \operatorname{Im} \langle \hat{C}(k), k \times \hat{X}(k) \rangle$  and neither a frame nor a normalisation is needed to evaluate it. Writing  $S_k$  for the matrix of  $A_k^{1/2}$ , the mode- $k$  term of Definition 3.2 is  $\langle S_k \hat{C}(k), i|k|JS_k \hat{X}(k) \rangle$ , whose inner matrix is  $S_k JS_k = \overline{\det(S_k)} J$  by Lemma 4.1. Because  $X$  and  $C(X, X)$  are real-valued vector fields,  $\hat{C}(-k) = \hat{C}(k)$  and  $\hat{X}(-k) = \hat{X}(k)$ , so the terms at  $k$  and  $-k$  are complex conjugates and their sum is twice a real part; the stated normalisation collects it. In particular  $D(X; A)$  is real, as it must be. Finally  $\det(S_k) = \det(A_k^{1/2}) = \sqrt{\det A_k}$ .  $\square$

## 5. THE CRITERION

Fix a finite family  $X_1, \dots, X_N$  of test fields and set  $M(C) := [c_k(X_s)]_{s \leq N, k \in \mathcal{P}}$ .

**Theorem 5.1.** *If  $C = B_A$  for a symmetric positive definite  $A$ , then the vector  $(\sqrt{\det A_k})_{k \in \mathcal{P}}$  is an entrywise positive null vector of  $M(C)$ . Consequently, if  $M(C)$  has full column rank, or if every null vector of  $M(C)$  has a non-positive entry, then  $C$  is not an Euler–Arnold operator, for any inertia operator whatsoever.*

*Proof.* By Lemma 3.1,  $D(X; A) = 0$  for every  $X$ , so by Proposition 4.2 the vector  $(\sqrt{\det A_k})_k$  annihilates every row of  $M(C)$ . It is entrywise positive because  $A_k$  is positive definite on a two-dimensional space, so  $\det A_k > 0$ .  $\square$

**Theorem 5.2** (locality). *Let  $C$  be a cascade operator on a mode set  $\mathcal{S}$ , finite or not, and let  $\mathcal{S}_0 \subseteq \mathcal{S}$  be finite and closed under  $k \mapsto -k$ . If  $M(C)$  computed on  $\mathcal{S}_0$  has full rank, then  $C$  is not an Euler–Arnold operator on  $\mathcal{S}$ , for any positive definite  $A$ .*

*Proof.* Suppose  $C = B_A$  on  $\mathcal{S}$ , and let  $X$  be supported on  $\mathcal{S}_0$ . By Proposition 4.2,  $c_k(X)$  carries the factor  $\hat{X}(k)$ , so  $c_k(X) = 0$  for every  $k \notin \mathcal{S}_0$ ; and  $\hat{C}(p)$  is determined by the  $\mathcal{S}_0$ -modes alone, since the triad sum runs over  $k + l = -p$  with  $\hat{X}(k), \hat{X}(l)$  vanishing off  $\mathcal{S}_0$ . Hence  $0 = D(X; A) = \sum_{k \in \mathcal{S}_0} \sqrt{\det A_k} c_k(X)$  for every such  $X$ , so  $(\sqrt{\det A_k})_{k \in \mathcal{S}_0}$  is an entrywise positive null vector of  $M(C)$  on  $\mathcal{S}_0$ , contradicting full rank.  $\square$

So the finite computations of §6 are not a model of the question: they are certificates for it. Proposition 6.7 exhibits, in exact integer arithmetic, one window of 32 modes on which a cascade has full rank; by Theorem 5.2 that operator is excluded on the whole lattice. Combining with Theorem 7.1, and noting that restriction to a window is a surjective linear map, so that the preimage of a proper Zariski-closed set is again one:

**Corollary 5.3.** *Outside a proper Zariski-closed set of coefficients and profiles, no local cascade operator on the full lattice is an Euler–Arnold operator, for any positive definite inertia operator.*

Two features deserve emphasis. First, the criterion never refers to the species count, to the number of terms in a sum, or to invertibility of a frame matrix; it applies verbatim to sums of arbitrarily many cascade terms. Second, it fails in two distinct ways — full rank, and rank

deficiency with no positive null vector — and both occur in practice, the second being the one that catches Tao’s own operator.

## 6. MEASUREMENTS

All computations below are rank and residual computations in fixed-size linear algebra; no optimiser or search is used at any point. Each is validated on data known to satisfy the hypothesis before being applied.

**Proposition 6.1** (numerical). *The premise of Lemma 3.1 holds in the discretisation. Computing  $\dot{m} = -P(u \cdot \nabla m + (\nabla u)^\top m)$  in real space on a  $16^3$  grid with spectral derivatives,  $|\frac{d}{dt}H|/|H|$  is at most  $1.5 \times 10^{-14}$  for  $A = I$ ,  $|k|^{-2}I$ ,  $|k|^{3/2}I$  and three genuinely anisotropic symbols of the form  $a(|k|)P_k + b(|k|)P_k(n \otimes n)P_k$ . Computing instead through (1) directly, the defect of Definition 3.2 is  $2.5 \times 10^{-16}$  relative.*

**Proposition 6.2** (numerical). *Formula (4) is exact: over random cascades, random positive definite  $A$  and random test fields, the two sides of (4) agree to  $8.4 \times 10^{-16}$  relative.*

**Proposition 6.3** (numerical). *On the mode set  $|k| \leq 2$  with 16 leg pairs, and for four random anisotropic  $A_0$ , the matrix  $M(B_{A_0})$  has corank exactly one, its smallest singular value being  $10^{-16}$  relative; the null vector agrees with  $(\sqrt{\det A_{0,k}})_k$  to ten decimal places, and its smallest entry is between 0.04 and 0.19 of its largest. It agrees with  $(\det A_{0,k})_k$  only to 0.972, so the computation distinguishes the two and selects the exponent predicted by Proposition 4.2.*

**Proposition 6.4** (numerical). *Let  $C$  be a cascade operator with  $\alpha$  admissible in the sense of Lemma 2.2 and random profiles. Then  $\langle C(X, X), X \rangle$  vanishes to  $10^{-17}$ , while  $D(X; A)$  is of order  $10^{-2}$  relative for every  $A$  tested, and  $M(C)$  has full column rank. Over twelve samples at  $m = 2, 3, 4$  the ratio  $\sigma_{\min}/\sigma_{\max}$  of  $M(C)$  lies between 0.047 and 0.22. Restricting the profiles to disjoint species supports, as in Tao’s construction, the rank is still full over eight further samples at  $m = 3, 4$ , with  $\sigma_{\min}/\sigma_{\max}$  between  $4.0 \times 10^{-3}$  and  $9.2 \times 10^{-2}$ .*

**Proposition 6.5** (numerical). *The conclusion is not an artefact of the mode set. On  $|k| \leq 2$ ,  $|k| \leq 2.5$  and  $|k| \leq 3$  — 16, 40 and 61 leg pairs — and on the images of the first two under a random invertible linear map, which preserves every triad while changing every length, angle and perpendicular plane, the corank of  $M(B_{A_0})$  is one in all five cases with null vector  $(\sqrt{\det A_{0,k}})_k$  positive, and  $M(C)$  has full rank in all five.*

**Proposition 6.6** (numerical). *For Tao’s own coefficients the criterion excludes by its second clause. On the frequency support only the four constants with  $\mu = 0$  and species  $\{1, 3, 4\}$  survive Paper I, Theorem 6.3, and among the orderings  $\alpha_{1,4,3} = \alpha_{4,1,3} = 0$ . His triads are disjoint, each mode lying in exactly one, so (4) decouples into three unknowns per triad. The cyclic symmetrisation of  $\alpha$  vanishes exactly, as Lemma 2.2 requires; the rank of the resulting 3-column matrix is 2, and the null vector is  $(0, 1, 0)$  — supported on the leg whose block is the vanishing one. Two of the three determinants are forced to zero, so no positive definite  $A$  exists.*

**Proposition 6.7** (exact). *There is a cascade operator, on an explicit integer mode set, for which  $M(C)$  has full rank over  $\mathbb{Q}$ , with no floating point involved. Take the integer lattice  $0 < |k|^2 \leq 4$  (32 modes, 16 pairs, 426 ordered triads); on each pair an integer basis  $b_1 = k \times e$ ,  $b_2 = k \times b_1$  of  $k^\perp$ ; integer profiles and Gaussian-integer test fields in that basis; and integer  $\alpha$  with vanishing cyclic symmetrisation. By Proposition 4.2 in the form  $c_k(X) = -2\text{Im}\langle \hat{C}(k), k \times \hat{X}(k) \rangle$ , every entry of  $M(C)$  is then an integer. For the data recorded in the accompanying code, Gaussian elimination over  $\mathbb{Q}$  gives  $\text{rank } M(C) = 16$ .*

*Remark 6.8.* Proposition 6.6 is a consistency check rather than news about Tao: his surviving coupling has a block that vanishes identically, and Paper II, Corollary 5.6 already excludes it on that ground. Its value is that his coefficients are exactly the kind of non-generic point at which the criterion might have gone blind, and it does not.

**Proposition 6.9** (numerical). *No cascade in the exceptional locus of Theorem 7.1 escapes by the second clause of Theorem 5.1, at least generically. Fix profiles and a vector  $d$  of positive determinants. Since  $\hat{C}$  is linear in  $\alpha$ , the set  $H(d) = \{\alpha : \sum_k d_k c_k(X) = 0 \text{ for all } X\}$  is a linear subspace of the admissible space, computable by rank. Let  $N = \{\alpha : C \equiv 0\}$ , the trivial directions. Over the tested profiles and several  $d$ , including  $d = (1, \dots, 1)$ , one finds  $H(d) = N$  exactly: in the cyclic-free space both have dimension 20, and in the admissible space of Lemma 2.2, where (4.2) is imposed, both are zero. So every admissible  $\alpha$  whose cascade conserves the momentum helicity for some positive definite  $A$  is the zero tensor.*

**Proposition 6.10** (numerical). *The exceptional locus admits a complete description in the tested setting, and it does not shelter anything. Since  $c_k$  is linear in  $\alpha$ , the matrix  $M(\alpha)$  is a linear pencil and  $Z$  is its rank-deficiency locus, stratified by the support of the null vector. Write  $N = \{\alpha : C \equiv 0\}$ . Then:*

- (1) *A null vector supported on a single leg  $k$  means  $c_k \equiv 0$ , which by Proposition 4.2 —  $\hat{X}(k)$  being free — happens exactly when the cascade has no output at leg  $k$ . For generic profiles that locus has dimension 4 inside the 20-dimensional admissible space, and two legs can be silent at once only for  $\alpha = 0$ .*
- (2) *For every other support,  $H(d) = N$ . Exhaustively over all 16, 120 and 560 supports of sizes 1, 2, 3 with three coefficient choices each, and over 3000 dense random  $d$ : for generic profiles  $\dim H(d) = 0$  except at the single-leg supports; for disjoint species supports  $\dim H(d)$  is larger, but every  $\alpha$  it contains gives the zero operator, to  $2 \times 10^{-14}$ .*
- (3) *On a branch,  $M(\alpha)$  has rank exactly 15 of 16, and its null space is one-dimensional and supported on that one leg — so it has zero entries and is not entrywise positive.*

**Corollary 6.11.** *In the tested setting no admissible cascade operator with  $C \neq 0$  is an Euler–Arnold operator — not merely the generic ones. Off  $Z$  there is no null vector at all; on  $Z$  the null vector is a coordinate direction, which fails positivity; and the  $\alpha$  admitting a positive null vector are exactly those with  $C \equiv 0$ .*

*Remark 6.12.* This is what happens at Tao’s own coefficients, Proposition 6.6: they lie on the branch belonging to the leg whose block vanishes, the null vector is that coordinate direction, and positivity fails. The mechanism there was not special to him.

*Remark 6.13.* Since  $\dim H(d)$  is upper semicontinuous in  $(\psi, d)$ , the equality  $H(d) = N$  at one point gives it on a Zariski-open dense set, by the argument of Theorem 7.1. What is not excluded is a non-generic pair  $(\psi, d)$  at which  $H(d)$  jumps; the locus of such pairs is closed and we have not described it.

## 7. WHAT THIS ESTABLISHES

**Theorem 7.1.** *Fix the integer mode set and the test fields of Proposition 6.7, and let the species count be four. Then there is a proper Zariski-closed subset  $Z$  of the admissible pairs  $(\alpha, \psi)$  such that for every  $(\alpha, \psi) \notin Z$  the associated cascade operator is not an Euler–Arnold operator, for any positive definite inertia operator.*

*Proof.* For fixed test fields,  $\hat{C}(k)$  is bilinear in the profile pairings and linear in  $\alpha$ , so each  $c_k(X_s)$  is a polynomial in  $(\alpha, \psi)$ . Full rank is the non-vanishing of some maximal minor, a polynomial

condition, so  $Z := \{\text{rank } M(C) < 16\}$  is Zariski-closed. It is proper because Proposition 6.7 exhibits a point outside it, by an exact computation over  $\mathbb{Q}$ . Off  $Z$  the matrix  $M(C)$  has full rank, so it has no null vector at all, and Theorem 5.1 applies.  $\square$

The witness is exact, so Theorem 7.1 carries no numerical input at all: the negative answer for generic cascade operators is a theorem, not a statement about a sample, and unlike the dimension counts of Paper III it does not degrade with the species count.

What is *not* established is threefold, and we state it plainly.

First, the exceptional locus of Theorem 7.1 is inhabited: Proposition 6.6 exhibits a point of it, where the criterion still excluded, by positivity rather than by rank. A cascade with both a rank deficiency and an entrywise positive null vector would escape the criterion entirely, and Propositions 6.9 and 6.10 say there is none: the  $\alpha$  admitting a positive null vector are exactly those giving the zero operator. What is measured rather than proved is the stratification itself, and it was checked on two profile families and one window.

Second, every computation reported here abstracts Tao's species balls  $B_i$  to their centres, so that each species contributes one wavevector and one profile direction. This is the same abstraction under which Paper I–Paper III operate, but it is an abstraction: his  $\psi_i$  genuinely span balls of positive radius.

Third, the mode sets are finite, but by Theorem 5.2 this is no longer a restriction on the conclusion: a full-rank window certifies the operator on any ambient set containing it, the full lattice included. What is not addressed is a cascade whose profiles are not of the Fourier form of Definition 2.1, and the continuum limit in which the number of species or the annulus geometry is allowed to vary with the scale.

*Remark 7.2.* The contrast with Paper III is worth recording. There the two classes were separated by a codimension that grew quadratically in the number of legs and still proved nothing, because the images might meet in special position. Here the separation is a rank, and a rank is not a dimension count: full rank is an open condition, it is witnessed by one evaluation, and Proposition 7.1 converts the witness into a statement about almost every cascade. The algebraic geometry that Paper III asked for turns out to be one line of it.

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The code underlying the computational statements is available from the author.

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